

Standardized Documentation Framework for MATLAB Scripts

Author: [Your Name]

Purpose: To maintain consistency, clarity, and ease of understanding for both current and future users of the code.

Documentation Guidelines for MATLAB Scripts

Inside the Function: Short & Clear Comments (Minimal Clutter)

- Each function should have a brief description at the top.
- Inputs and outputs should be listed concisely.
- Core steps should be briefly commented to aid readability.

Example: Standard Function Header

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```
function wingMesh = generateWingMesh(sparPosition, ribSpacing,  
stringerDistribution)
```

```
% Generates the structural mesh of the wing.
```

```
%
```

```
% Inputs:
```

```
% - sparPosition: Spar location as a percentage of chord.
```

```
% - ribSpacing: Distance between ribs in meters.
```

```
% - stringerDistribution: Distribution of stringers along the span.
```

```
%
```

```
% Output:
```

```
% - wingMesh: Struct containing the nodal and element data.
```

```
% Compute the rib locations
```

```
ribLocations = 0:ribSpacing:10; % Example: fixed span length of 10m
```

```
% Generate structural components
```

```

sparNodes = generateSparNodes(sparPosition, ribLocations);

stringerNodes = generateStringerNodes(stringerDistribution, ribLocations);

% Merge into a single structure
wingMesh.nodes = [sparNodes; stringerNodes];

end

```

Add a Note at the Top Indicating More Details Below

- If a function has **complex logic**, users should know **where to find a deeper explanation**.
- Place a **single-line reference** at the beginning of the file:

matlab

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%  Detailed comments available at the bottom of this script.

At the Bottom: Full Detailed Explanation

- **Purpose:** Explains the function in depth for users unfamiliar with the project.
- **Includes:**
 - A summary of what the function does.
 - A step-by-step breakdown of key calculations.
 - Important assumptions, constraints, or limitations.
 - Any modification notes or potential improvements.

Example: Standardized Bottom Documentation Section

matlab

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%% =====

%  **Detailed Explanation of `generateWingMesh`**

```

% =====

%

% Purpose:

% This function generates the structural mesh of the wing, considering:

% - Spar Locations ( `sparPosition` ) → Defines the position of main spars as a
% of the chord.

% - Rib Spacing ( `ribSpacing` ) → Determines the structural divisions along the
span.

% - Stringer Distribution ( `stringerDistribution` ) → Defines stringer positioning.

%

% Workflow:

% 1 Compute Rib Locations → Uses a fixed span length of 10m (adjustable
later).

% 2 Generate Spar Nodes → Calls `generateSparNodes()` to determine spar
placements.

% 3 Generate Stringer Nodes → Uses `generateStringerNodes()` for distributed
stringers.

% 4 Merge Nodes → Combines all elements into a structured output
( `wingMesh` ).

%

% Assumptions & Constraints:

% - The function assumes a fixed wing span of 10m, but this can be changed.

% - `sparPosition` values must be in `[0,1]` to ensure valid placement.

%

% Modification Notes:

% - If the wing span is variable, replace `10m` with an input parameter.

% - Ensure that `sparPosition` values are validated before computation.

%

% =====

```

📄 Ensure Consistency Across All Scripts

- Every function should follow this format:
 - ✅ Short **header comments** inside the function.
 - ✅ A "**More details at the bottom**" reference at the top (if needed).
 - ✅ A **detailed documentation section** at the bottom of the script.
 - If a function is **simple**, the detailed section is **optional**.
-

🎯 Why This Approach is the Best?

- ✅ **Efficiency** → The function remains **readable and uncluttered**.
 - ✅ **Clarity** → Future users can quickly **find more information** without disrupting code flow.
 - ✅ **Flexibility** → Works for **both simple and complex** functions.
 - ✅ **Consistency** → Ensures **uniform documentation** across all scripts.
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📝 Final Note

This **documentation framework** will guide us in maintaining high-quality, well-documented scripts **without sacrificing efficiency**. 🚀 If the project scales or more contributors join, they can **easily understand the code without reading the thesis**.